Solutions

8.9 (a) $\bar{X} \pm Z \cdot \frac{\sigma}{\sqrt{n}} = 0.995 \pm 2.58 \cdot \frac{0.02}{\sqrt{50}}$ $0.9877 \le \mu \le 1.0023$

(b) Since the value of 1.0 is included in the interval, there is no reason to believe that the mean is different from 1.0 gallon.

(c) No. Since σ is known and $n = 50$, from the Central Limit Theorem, we may assume that the sampling distribution of $\bar{X}$ is approximately normal.

8.9 (d) The reduced confidence level narrows the width of the confidence interval.

cont. $\bar{X} \pm Z \cdot \frac{\sigma}{\sqrt{n}} = 0.995 \pm 1.96 \cdot \frac{0.02}{\sqrt{50}}$ $0.9895 \le \mu \le 1.0005$

(b) Since the value of 1.0 is still included in the interval, there is no reason to believe that the mean is different from 1.0 gallon.

8.10 (a)

Confidence Interval Estimate for the Mean	
Data	
Population Standard Deviation	**1000**
Sample Mean	**7500**
Sample Size	**64**
Confidence Level	**95%**
Intermediate Calculations	
Standard Error of the Mean	125
Z Value	-1.9600
Interval Half Width	244.9955
Confidence Interval	
Interval Lower Limit	**7255.00**
Interval Upper Limit	**7745.00**

$\bar{X} \pm Z \cdot \frac{\sigma}{\sqrt{n}} = 7500 \pm 1.96 \cdot \frac{1000}{\sqrt{64}}$ $7255.00 \le \mu \le 7745.00$

(b) No. The manufacturer cannot support a claim that the bulbs last an average 8,000 hours with a 95% level of confidence because 8,000 does not fall inside the 95% confidence interval.

(c) No. Since σ is known and $n = 64$, from the Central Limit Theorem, we may assume that the sampling distribution of $\bar{X}$ is approximately normal.

(d) The confidence interval is narrower based on a process standard deviation of 800 hours rather than the original assumption of 1000 hours.

(a) $\bar{X} \pm Z \cdot \frac{\sigma}{\sqrt{n}} = 7500 \pm 1.96 \cdot \frac{800}{\sqrt{64}}$ $7304.00 \le \mu \le 7696.00$

(b) No. The manufacturer cannot support a claim that the bulbs last an average 8,000 hours with a 95% level of confidence because 8,000 still does not fall inside the 95% confidence interval.

8.15 (a) PHStat output:

Confidence Interval Estimate for the Mean

Data	
Sample Standard Deviation	**200**
Sample Mean	**1500**
Sample Size	**100**
Confidence Level	**95%**

Intermediate Calculations	
Standard Error of the Mean	20
Degrees of Freedom	99
t Value	1.9842
Interval Half Width	39.6843

Confidence Interval	
Interval Lower Limit	**1460.32**
Interval Upper Limit	**1539.68**

$$\bar{X} \pm t \cdot \frac{S}{\sqrt{n}} = 1500 \pm 1.9842 \cdot \frac{200}{\sqrt{100}} \qquad \$\ 1460.32 \le \mu \le \$\ 1539.68$$

(b) You can be 95% confident that the population mean spending for all Amazon Prime member shoppers is somewhere between $1460.32 and $1539.68.

8.16 (a) $$\bar{X} \pm t \cdot \frac{S}{\sqrt{n}} = 82 \pm 1.9890 \cdot \frac{9}{\sqrt{84}} \qquad \$80.05 \le \mu \le \$83.95$$

PHStat output:

Confidence Interval Estimate for the Mean

Data	
Sample Standard Deviation	**9**
Sample Mean	**82**
Sample Size	**84**
Confidence Level	**95%**

Intermediate Calculations	
Standard Error of the Mean	0.981980506
Degrees of Freedom	83
t Value	1.9890
Interval Half Width	1.9531

Confidence Interval	
Interval Lower Limit	**80.05**
Interval Upper Limit	**83.95**

(b) You can be 95% confident that the population mean one-time gift donation is somewhere between $80.05 and $83.95.

8.24 (a) PHStat output:

Confidence Interval Estimate for the Mean

Data	
Sample Standard Deviation	**6.651868497**
Sample Mean	**16.39285714**
Sample Size	**28**
Confidence Level	**95%**

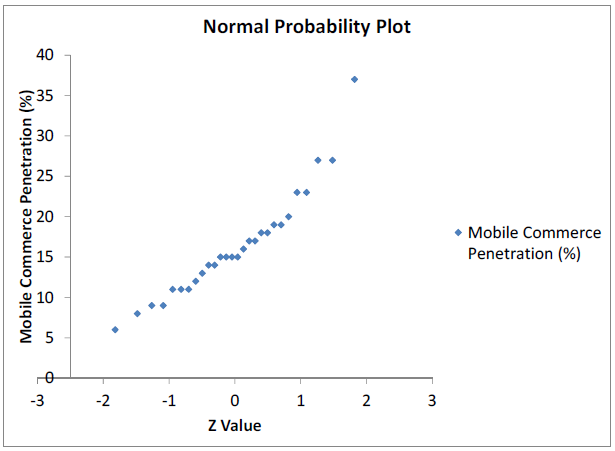
Intermediate Calculations	
Standard Error of the Mean	1.257084985
Degrees of Freedom	27
t Value	2.0518
Interval Half Width	2.5793

Confidence Interval	
Interval Lower Limit	**13.81**
Interval Upper Limit	**18.97**

$$\bar{X} \pm t\frac{S}{\sqrt{n}} = 16.3929 \pm 2.0518\left(\frac{6.6519}{\sqrt{28}}\right) \qquad 13.81 \le \mu \le 18.97$$

(b) The population distribution needs to be normally distributed.

(c)



The normal probability plot suggests a right skewed distribution due to the outlier from the Republic of Korea.

Hypothesis Testing

Solutions:

1. a. H_0: $\mu \leq 600$ Manager's claim.
   H_a: $\mu > 600$

   b. We are not able to conclude that the manager's claim is wrong.

   c. The manager's claim can be rejected. We can conclude that $\mu > 600$.

2. a. H_0: $\mu \leq 14$
   H_a: $\mu > 14$ Research hypothesis

   b. There is no statistical evidence that the new bonus plan increases sales volume.

   c. The research hypothesis that $\mu > 14$ is supported. We can conclude that the new bonus plan increases the mean sales volume.

3. a. H_0: $\mu = 32$ Specified filling weight
   H_a: $\mu \neq 32$ Overfilling or underfilling exists

   b. There is no evidence that the production line is not operating properly. Allow the production process to continue.

   c. Conclude $\mu \neq 32$ and that overfilling or underfilling exists. Shut down and adjust the production line.

4. a. H_0: $\mu \geq 220$
   H_a: $\mu < 220$ Research hypothesis to see if mean cost is less than \$220.

   b. We are unable to conclude that the new method reduces costs.

   c. Conclude $\mu < 220$. Consider implementing the new method based on the conclusion that it lowers the mean cost per hour.

5. a. Conclude that the population mean monthly cost of electricity in the Chicago neighborhood is greater than \$104 and hence higher than in the comparable neighborhood in Cincinnati.

   b. The Type I error is rejecting H_0 when it is true. This error occurs if the researcher concludes that the population mean monthly cost of electricity is greater than \$104 in the Chicago neighborhood when the population mean cost is actually less than or equal to \$104.

c. The Type II error is accepting H_0 when it is false. This error occurs if the researcher concludes that the population mean monthly cost for the Chicago neighborhood is less than or equal to \$104 when it is not.

6. a. H_0: $\mu \leq 1$ The label claim or assumption.
H_a: $\mu > 1$

b. Claiming $\mu > 1$ when it is not. This is the error of rejecting the product's claim when the claim is true.

c. Concluding $\mu \leq 1$ when it is not. In this case, we miss the fact that the product is not meeting its label specification.

7. a. H_0: $\mu \leq 8000$
H_a: $\mu > 8000$ Research hypothesis to see if the plan increases average sales.

b. Claiming $\mu > 8000$ when the plan does not increase sales. A mistake could be implementing the plan when it does not help.

c. Concluding $\mu \leq 8000$ when the plan really would increase sales. This could lead to not implementing a plan that would increase sales.

8. a. H_0: $\mu \geq 220$
H_a: $\mu < 220$

b. Claiming $\mu < 220$ when the new method does not lower costs. A mistake could be implementing the method when it does not help.

c. Concluding $\mu \geq 220$ when the method really would lower costs. This could lead to not implementing a method that would lower costs.

9.14 (a) PHStat output:

Z Test of Hypothesis for the Mean	
Data	
Null Hypothesis μ=	**7500**
Level of Significance	**0.05**
Population Standard Deviation	**1000**
Sample Size	**64**
Sample Mean	**7250**
Intermediate Calculations	
Standard Error of the Mean	125.0000
***Z* Test Statistic**	**-2.0000**
Two-Tail Test	
Lower Critical Value	**-1.9600**
Upper Critical Value	**1.9600**
***p*-Value**	**0.0455**
Reject the null hypothesis	

H_0: $\mu = 7500$. The mean life of a large shipment of light bulbs is equal to 7500 hours.
H_1: $\mu \neq 7500$. The mean life of a large shipment of light bulbs differs from 7500 hours.
Decision rule: Reject H_0 if $|Z_{STAT}| > 1.96$

Test statistic: $Z_{STAT} = \dfrac{\bar{X} - \mu}{\sigma / \sqrt{n}} =$ -2.0000

Decision: Since $|Z_{STAT}| > 1.96$, reject H_0. There is enough evidence to conclude that the mean life of a large shipment of light bulbs differs from 7500 hours.

(b) *p*-value = 0.0455. If the population mean life of a large shipment of light bulbs is indeed equal to 7500 hours, the probability of obtaining a test statistic that is more than 2.0 standard error units away from 0 is 0.0455.

9.14 cont. (c) PHStat output:

Confidence Interval Estimate for the Mean	
Data	
Population Standard Deviation	**1000**
Sample Mean	**7250**
Sample Size	**64**
Confidence Level	**95%**
Intermediate Calculations	
Standard Error of the Mean	125.0000
Z Value	-1.9600
Interval Half Width	244.9955
Confidence Interval	
Interval Lower Limit	**7005.00**
Interval Upper Limit	**7495.00**

(c) $\bar{X} \pm Z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}} = 7250 \pm 1.96\dfrac{1000}{\sqrt{64}}$ $\qquad 7005.00 \leq \mu \leq 7495.00$

(d) You are 95% confident that the population mean life of a large shipment of light bulbs is somewhere between 7005.00 and 7495.00 hours.
Since the 95% confidence interval does not contain the hypothesized value of 7500, you will reject H_0. The conclusions are the same.

9.16 (a) PHStat output:

Data	
Null Hypothesis μ =	**1**
Level of Significance	**0.01**
Population Standard Deviation	**0.02**
Sample Size	**50**
Sample Mean	**0.995**
Intermediate Calculations	
Standard Error of the Mean	0.002828427
Z Test Statistic	**-1.767766953**
Two-Tail Test	
Lower Critical Value	**-2.575829304**
Upper Critical Value	**2.575829304**
***p*-Value**	**0.077099872**
Do not reject the null hypothesis	

H_0: $\mu = 1$. The mean amount of water is 1 gallon.

H_1: $\mu \neq 1$. The mean amount of water differs from 1 gallon.

Decision rule: Reject H_0 if $|Z_{STAT}| > 2.5758$

(a) Test statistic: $Z_{STAT} = \dfrac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \dfrac{.995 - 1}{.02/\sqrt{50}} = -1.7678$

Decision: Since $|Z_{STAT}| < 2.5758$, do not reject H_0. There is not enough evidence to conclude that the mean amount of water contained in 1-gallon bottles purchased from a nationally known water bottling company is different from 1 gallon.

(b) *p*-value = 0.0771. If the population mean amount of water contained in 1-gallon bottles purchased from a nationally known water bottling company is actually 1 gallon, the probability of obtaining a test statistic that is more than 1.7678 standard error units away from 0 is 0.0771.

(c) PHStat output:

Data	
Population Standard Deviation	**0.02**
Sample Mean	**0.995**
Sample Size	**50**
Confidence Level	**99%**
Intermediate Calculations	
Standard Error of the Mean	0.002828427
Z Value	-2.5758293
Interval Half Width	0.007285545
Confidence Interval	
Interval Lower Limit	**0.987714455**
Interval Upper Limit	**1.002285545**

(c) $\bar{X} \pm Z_{a/2}\dfrac{\sigma}{\sqrt{n}} = .995 \pm 2.5758\dfrac{.02}{\sqrt{50}}$ $\qquad 0.9877 \leq \mu \leq 1.0023$

You are 99% confident that population mean amount of water contained in 1-gallon bottles purchased from a nationally known water bottling company is somewhere between 0.9877 and 1.0023 gallons.

(d) Since the 99% confidence interval does contain the hypothesized value of 1, you will not reject H_0. The conclusions are the same.

22. a. H_0: $\mu = 8$

H_a: $\mu \neq 8$

b. $z = \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \dfrac{8.4 - 8.0}{3.2 / \sqrt{120}} = 1.37$

Because $z > 0$, *p*-value is two times the upper tail area

Using normal table with $z = 1.37$: p-value = 2(1 - .9147) = .1706

c. p-value > .05; do not reject H_0. Cannot conclude that the population mean waiting time differs from 8 minutes.

d. $\bar{x} \pm z_{.025}(\sigma/\sqrt{n})$

$8.4 \pm 1.96(3.2/\sqrt{120})$

$8.4 \pm .57$ (7.83 to 8.97)

Yes; $\mu = 8$ is in the interval. Do not reject H_0.

9.24 PHStat output:

t Test for Hypothesis of the Mean	
Data	
Null Hypothesis μ =	**3.7**
Level of Significance	**0.05**
Sample Size	**64**
Sample Mean	**3.57**
Sample Standard Deviation	**0.8**
Intermediate Calculations	
Standard Error of the Mean	0.1
Degrees of Freedom	63
***t* Test Statistic**	**-1.3**
Two-Tail Test	
Lower Critical Value	**-1.9983405**
Upper Critical Value	**1.9983405**
***p*-Value**	**0.1983372**
Do not reject the null hypothesis	

(a) $H_0 : \mu = 3.7 \quad H_1 : \mu \neq 3.7$

Decision rule: Reject H_0 if $|t_{STAT}| > 1.9983 \quad d.f. = 63$

Test statistic: $t_{STAT} = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} = \dfrac{3.57 - 3.7}{0.8/\sqrt{64}} = -1.3$

Decision: Since $|t_{STAT}| < 1.9983$, do not reject H_0. There is not enough evidence to conclude that the population mean waiting time is different from 3.7 minutes at the 0.05 level of significance.

(b) The sample size of 64 is large enough to apply the Central Limit Theorem and, hence, you do not need to be concerned about the shape of the population distribution when conducting the t-test in (a). In general, the t test is appropriate for this sample size except for the case where the population is extremely skewed or bimodal.

9.26 PHStat output:

t Test for Hypothesis of the Mean	
Data	
Null Hypothesis μ=	**1475**
Level of Significance	**0.05**
Sample Size	**100**
Sample Mean	**1500**
Sample Standard Deviation	**200**
Intermediate Calculations	
Standard Error of the Mean	20.0000
Degrees of Freedom	99
***t* Test Statistic**	**1.2500**
Two-Tail Test	
Lower Critical Value	**-1.9842**
Upper Critical Value	**1.9842**
***p*-Value**	**0.2142**
Do not reject the null hypothesis	

(a) $H_0 : \mu = \$1,475$ $H_1 : \mu \neq \$1,475$

Decision rule: Reject H_0 if p-value < 0.05

Test statistic: $t_{STAT} = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} = 1.2500$

p-value = 0.2142

Decision: Since the p-value of 0.2142 > 0.05, do not reject H_0. There is not enough evidence to conclude that the population mean amount spent on Amazon.com by Amazon Prime member shoppers is different from $1,475.

(b) The p-value is 0.2142. If the population mean is indeed $1,475, the probability of obtaining a test statistic that is more than 1.25 standard error units away from 0 in either direction is 0.2142.

9.28 PHStat output:

t Test for Hypothesis of the Mean	
Data	
Null Hypothesis μ=	**6.5**
Level of Significance	**0.05**
Sample Size	**15**
Sample Mean	**7.09**
Sample Standard Deviation	**1.406031226**
Intermediate Calculations	
Standard Error of the Mean	0.3630
Degrees of Freedom	14
***t* Test Statistic**	**1.6344**
Two-Tail Test	
Lower Critical Value	**-2.1448**
Upper Critical Value	**2.1448**
***p*-Value**	**0.1245**
Do not reject the null hypothesis	

(a) $H_0 : \mu = \$6.50 \quad H_1 : \mu \neq \$6.50$

Decision rule: Reject H_0 if $|t_{STAT}| > 2.1448$ or p-value < 0.05

Test statistic: $t_{STAT} = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} = 1.6344$

Decision: Since $|t_{STAT}| < 2.1448$, do not reject H_0. There is not enough evidence to conclude that the mean amount spent for lunch is different from $6.50.

(b) The p-value is 0.1245. If the population mean is indeed $6.50, the probability of obtaining a test statistic that is more than 1.6344 standard error units away from 0 in either direction is 0.4069.

(c) That the distribution of the amount spent on lunch is normally distributed.

(d) With a sample size of 15, it is difficult to evaluate the assumption of normality. However, the distribution may be fairly symmetric because the mean and the median are close in value. Also, the boxplot appears only slightly skewed so the normality assumption does not appear to be seriously violated.

Chi-Square Test

11. a. H_0: Type of ticket purchased is independent of the type of flight

H_a: Type of ticket purchased is not independent of the type of flight

Expected Frequencies:

$e_{11} = 35.59 \quad e_{12} = 15.41$

$e_{21} = 150.73 \quad e_{22} = 65.27$

$e_{31} = 455.68 \quad e_{32} = 197.32$

Ticket	Flight	Observed Frequency (f_i)	Expected Frequency (e_i)	Chi–square $(f_i - e_i)^2 / e_i$
First	Domestic	29	35.59	1.22
First	International	22	15.41	2.82
Business	Domestic	95	150.73	20.61
Business	International	121	65.27	47.59
Full Fare	Domestic	518	455.68	8.52
Full Fare	International	135	197.32	19.68
	Totals:	920		$\chi^2 = 100.43$

Degrees of freedom = $(r - 1)(c - 1) = (3 - 1)(2 - 1) = 2$

Using the χ^2 table with $df = 2$, $\chi^2 = 100.43$ shows the p–value is less than .005.

Using Excel or Minitab, the p–value corresponding to $\chi^2 = 100.43$ is .0000.

p–value $\leq$ .05, reject H_0. Conclude that the type of ticket purchased is not independent of the type of flight. We can expect the type of ticket purchased to depend upon whether the flight is domestic or international.

b. Column Percentages

	Type of Flight	
Type of Ticket	Domestic	International
First Class	4.5%	7.9%
Business Class	14.8%	43.5%
Economy Class	80.7%	48.6%

A higher percentage of first class and business class tickets are purchased for international flights compared to domestic flights. Economy class tickets are purchased more for domestic flights. The first class or business class tickets are purchased for more than 50% of the international flights; 7.9% + 43.5% = 51.4%.

12. a. H_0: Employment plan is independent of the type of company

H_a: Employment plan is not independent of the type of company

Observed Frequency (f_{ij})

Employment Plan	Private	Public	Total
Add Employees	37	32	69
No Change	19	34	53
Lay Off Employees	16	42	58
Total	72	108	180

Expected Frequency (e_{ij})

Employment Plan	Private	Public	Total
Add Employees	27.6	41.4	69
No Change	21.2	31.8	53
Lay Off Employees	23.2	34.8	58
Total	72.0	108.0	180

Chi Square Calculations $(f_{ij} - e_{ij})^2 / e_{ij}$

Employment Plan	Private	Public	Total
Add Employees	3.20	2.13	5.34
No Change	0.23	0.15	0.38
Lay Off Employees	2.23	1.49	3.72
		χ^2 =	9.44

Degrees of freedom = $(r - 1)(c - 1) = (3 - 1)(2 - 1) = 2$

Using the χ^2 table with $df = 2$, $\chi^2 = 9.44$ shows the p–value is less than .01

Using Excel or Minitab, the p–value corresponding to $\chi^2 = 9.44$ is .0089

p–value $\leq$ .05, reject H_0. Conclude the employment plan is not independent of the type of company. Thus, we expect employment plan to differ for private and public companies.

b. Column probabilities – For example, 37/72 = .5139

Employment Plan	Private	Public
Add Employees	.5139	.2963
No Change	.2639	.3148
Lay Off Employees	.2222	.3889

Employment opportunities look to be much better for private companies with over 50% of private companies planning to add employees (51.39%). Public companies have the greater proportions of no change and lay off employees planned. 38.89% of public companies are planning to lay off employees over the next 12 months. 69/180 = .3833, or 38.33% of the companies in the survey are planning to hire and add employees during the next 12 months.

14. a. H_0: Quality rating is independent of the education of the owner

H_a: Quality rating is not independent of the education of the owner

Observed Frequencies (f_{ij})

Quality Rating	Some HS	HS Grad	Some College	College Grad	Total
Average	35	30	20	60	145

Outstanding	45	45	50	90	230
Exceptional	20	25	30	50	125
Total	100	100	100	200	500

Expected Frequencies (e_{ij})

Quality Rating	Some HS	HS Grad	Some College	College Grad	Total
Average	29	29	29	58	145
Outstanding	46	46	46	92	230
Exceptional	25	25	25	50	125
Total	100	100	100	200	500

Chi Square Calculations $(f_{ij} - e_{ij})^2 / e_{ij}$

Quality Rating	Some HS	HS Grad	Some College	College Grad	Total
Average	1.24	.03	2.79	.07	4.14
Outstanding	.02	.02	.35	.04	.43
Exceptional	1.00	.00	1.00	.00	2.00
					$\chi^2 = 6.57$

Degrees of freedom = $(r - 1)(c - 1) = (3 - 1)(4 - 1) = 6$

Using the χ^2 table with $df = 6$, $\chi^2 = 6.57$ shows the p–value is greater than .10

Using Excel or Minitab, the p–value corresponding to $\chi^2 = 6.57$ is .3624

p–value > .05, do not reject H_0. We are unable to conclude that the quality rating is not independent of the education of the owner. Thus, quality ratings are not expected to differ with the education of the owner.

b. Average: 145/500 = 29%

Outstanding: 230/500 = 46%

Exceptional: 125/500 = 25%

New owners look to be pretty satisfied with their new automobiles with almost 50% rating the quality outstanding and over 70% rating the quality outstanding or exceptional.

15. a. H_0: Quality of Management is independent of the Reputation of the Company

H_a: Quality of Management is not independent of the Reputation of the Company

Observed Frequencies (f_{ij})

Quality of Management	Excellent	Good	Fair	Total
Excellent	40	25	5	70
Good	35	35	10	80
Fair	25	10	15	50
Total	100	70	30	200

Expected Frequencies (e_{ij})

Quality of Management	Excellent	Good	Fair	Total
Excellent	35.0	24.5	10.5	70
Good	40.0	28.0	12.0	80
Fair	25.0	17.5	7.5	50
Total	100	70	30	200

Chi Square Calculations $(f_{ij} - e_{ij})^2 / e_{ij}$

Quality of Management	Excellent	Good	Fair	Total
Excellent	.71	.01	2.88	3.61
Good	.63	1.75	.33	2.71
Fair	.00	3.21	7.50	10.71
				$\chi^2 = 17.03$

Degrees of freedom = $(r - 1)(c - 1) = (3 - 1)(3 - 1) = 4$

Using the χ^2 table with $df = 4$, $\chi^2 = 17.03$ shows the p–value is less than .005

Using Excel or Minitab, the p–value corresponding to $\chi^2 = 17.03$ is .0019

p–value $\leq .05$, reject H_0. Conclude that the rating for the quality of management is not independent of the rating for the reputation of the company.

b. Using the highest column probabilities, if the reputation of the company is

Excellent: There is a 40/100 = .40 chance the quality of management will also be excellent.

Good: There is a 35/70 = .50 chance the quality of management will also be good.

Fair: There is a 15/30 = .50 chance the quality of management will also be fair.

The highest probabilities are that the two variables will have the same ratings. Thus, the two rating are associated.

17. a. H_0: Hours of sleep per night is independent of age

H_a: Hours of sleep per night is not independent of age

Observed Frequencies (f_{ij})

Hours of Sleep	39 or younger	40 or older	Total
Fewer than 6	38	36	74
6 to 6.9	60	57	117
7 to 7.9	77	75	152
8 or more	65	92	157
Total	240	260	500

Expected Frequencies (e_{ij})

Hours of Sleep	39 or younger	40 or older	Total
Fewer than 6	35.52	38.48	74
6 to 6.9	56.16	60.84	117
7 to 7.9	72.96	79.04	152
8 or more	75.36	81.64	157
Total	240	260	500

Chi Square Calculations $(f_{ij} - e_{ij})^2 / e_{ij}$

Hours of Sleep	39 or younger	40 or older	Total
Fewer than 6	.17	.16	.33
6 to 6.9	.26	.24	.50
7 to 7.9	.22	.21	.43
8 or more	1.42	1.31	2.74
			$\chi^2 = 4.01$

Degrees of freedom = $(r - 1)(c - 1) = (4 - 1)(2 - 1) = 3$

Using the χ^2 table with $df = 3$, $\chi^2 = 4.01$ shows the p–value is greater than .10.

Using Excel or Minitab, the p–value corresponding to $\chi^2 = 4.01$ is .2604.

p–value > .05, do not reject H_0. Cannot reject the assumption that age and hours of sleep are independent.

b. Since age does not appear to have an association on hours of sleep, use the overall row percentages.

Fewer than 6	74/500	= .148	14.8%
6 to 6.9	117/500	= .234	23.4%
7 to 7.9	152/500	= .304	30.4%
8 or more	157/500	= .314	31.4%

30.4% + 31.4% = 61.8% of individuals get seven or more hours of sleep a night.